Test Case Document

Project: Task Manager Application

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Revision History

Version	Date	Author	Change Summary
1.0	15-09-2026	Student Team	Sample test cases in Software Test Plan section 13.1; IDs in the RTM.
2.0	04-11-2026	Student Team	Every RTM test case written out in full and updated for the changes made in Sprints 1 and 2. TC-Val-01 added. Results for v1.0 recorded.
2.1	05-11-2026	Student Team	TC-Val-01 extended in Sprint 3. Results recorded for v1.0.1.

Approvals

Role	Name	Signature / Date
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1. Summary

30 test cases cover all 31 SRS requirements. Results of the latest run: **28 pass, 2 partial, 0 fail**.

Test Case	Requirement	Title	Result
TC-Auth-01	TMA-F-001	Register a new user with valid required information	**PASS**
TC-Auth-02	TMA-F-002	Log in with valid credentials	**PASS**
TC-Auth-03	TMA-F-003	Log in with invalid credentials	**PASS**
TC-Auth-04	TMA-F-004	Log out	**PASS**
TC-Task-01	TMA-F-010	Create a task with a title and optional description	**PASS**
TC-Task-02	TMA-F-011	Edit an existing task	**PASS**
TC-Task-03	TMA-F-012	Delete a task after confirmation	**PASS**
TC-Task-04	TMA-F-013	Mark a task complete and reopen it	**PASS**
TC-Sec-01	TMA-F-014, TMA-SR-003	Attempt to access another user's task	**PASS**
TC-Sec-02	TMA-SR-002, TMA-NF-003	Inspect stored password data and data protection	**PASS**
TC-Sec-03	TMA-SR-001	HTTPS / TLS for network communication	**PARTIAL**
TC-Sec-04	TMA-SR-004	Session tokens are protected and invalidated	**PASS**
TC-Sec-05	TMA-SR-005	Security events are logged without secrets	**PASS**

Test Case	Requirement	Title	Result
TC-Val-01	Test Plan 5.1, SAD 4.4	Basic input validation and fuzz style checks	**PASS**
TC-Org-01	TMA-F-020	Assign a priority to a task	**PASS**
TC-Org-02	TMA-F-021	Assign and update task status	**PASS**
TC-Org-03	TMA-F-022	Create or select categories / tags	**PASS**
TC-Org-04	TMA-F-023	Set or change a task due date	**PASS**
TC-Org-05	TMA-F-024	Configure a reminder	**PASS**
TC-Search-01	TMA-F-030	Search tasks by title or description keyword	**PASS**
TC-Search-02	TMA-F-031	Filter by status, priority and category	**PASS**
TC-Search-03	TMA-F-032	Sort by due date, priority or status	**PASS**
TC-Progress-01	TMA-F-033	Progress summary	**PASS**
TC-Notify-01	TMA-F-040	Notification generated for a due reminder	**PASS**
TC-Notify-02	TMA-F-041	Manage notification and task preferences	**PASS**
TC-Admin-01	TMA-F-042	Administrator user management	**PASS**
TC-Perf-01	TMA-NF-001	Load the dashboard under normal load	**PASS**
TC-Reliability-01	TMA-NF-002	System availability	**PARTIAL**
TC-Integrity-01	TMA-NF-004	Data consistency across operations	**PASS**
TC-UX-01	TMA-NF-005	Responsive and accessible user interface	**PASS**

2. What changed during the sprints

Deviation numbers refer to `docs/DEVIATIONS.md`.

Test Case	Change from Test Plan v1.0
TC-Auth-01	Updated. Sprint 1 fixed the password policy (8-128 characters, a letter and a digit) and the name limit (100). Registration now also logs the user straight in.
TC-Auth-02	Updated. Expected result now names the bearer session token chosen in Sprint 1 (SAD 4.6 left the mechanism open).
TC-Auth-03	Extended. Lockout steps added for the rate limit introduced in Sprint 2 (deviation D11).
TC-Auth-04	Unchanged in intent; written out in full (the ID only appeared in the RTM).
TC-Task-01	Updated. Length limits added. The form also sets priority, status, due date, categories and a reminder in the same step, in one transaction.
TC-Task-02	Extended. All-or-nothing validation and the change log (TaskHistory) were added during Sprint 1.
TC-Task-03	Extended. Cancel path and cascade removal now checked.
TC-Task-04	Unchanged in intent; written out in full.
TC-Sec-01	Extended. Now covers every route, and the log entry added in Sprint 2.
TC-Sec-02	Extended. Sprint 2 added security headers and SQL injection checks. Scrypt recorded as the approved mechanism.
TC-Sec-03	Changed. A trusted certificate belongs to the hosting environment, so this case verifies enforcement and a self-signed TLS connection (deviation D12).
TC-Sec-04	Unchanged in intent; written out in full.
TC-Sec-05	Unchanged in intent; written out in full. Administrators can also read events through the API.

Test Case	Change from Test Plan v1.0
TC-Val-01	NEW in Part 3. Test Plan 5.1 asks for these checks but no test case ID was assigned to them. Failed at v1.0 (DEF-01 to DEF-04). Sprint 3 fixed the defects and extended the case to sweep every route with a seeded fuzz test, which also found DEF-05.
TC-Org-01	Unchanged in intent; written out in full.
TC-Org-02	Changed. OVERDUE is derived from the due date rather than stored or chosen (deviation D4).
TC-Org-03	Extended. Case-insensitive reuse and per-user privacy added in Sprint 2.
TC-Org-04	Extended. The UTC storage and local display convention was fixed in Sprint 2.
TC-Org-05	Updated. The relative reminder options were defined in Sprint 2.
TC-Search-01	Unchanged from the Test Plan sample.
TC-Search-02	Updated. Overdue added as a filter value (follows D4).
TC-Search-03	Extended. Sorting by creation date and title, and paging, added in Sprint 2.
TC-Progress-01	Extended. Completion rate and 'due in the next 7 days' added in Sprint 2.
TC-Notify-01	Changed. In-app delivery is fully tested. The e-mail provider is a stub tested at its interface (deviation D9).
TC-Notify-02	Extended. Task preferences added, which the SAD class diagram did not model (deviation D10).
TC-Admin-01	Extended. Self-lockout rule and command-line creation of the first administrator added in Sprint 2.
TC-Perf-01	Changed. A scripted load test replaces JMeter (deviation D5). Part 3 adds a run over real HTTP.
TC-Reliability-01	Changed. The monthly figure can only be measured on a deployed system (deviation D13).
TC-Integrity-01	Extended. Part 3 adds the concurrent-writer and kill recovery scenarios from SRS section 6.
TC-UX-01	Extended. The manual Sprint 2 checklist is now automated in a real browser in Part 3.

3. Test cases

Test Case ID	TC-Auth-01: Register a new user with valid required information
Requirement	TMA-F-001
Module / Priority	Authentication / High
Preconditions	The email address is not yet registered.
Test data	Name, unique email, password of 8-128 characters with a letter and a digit.
Steps	1. Open the app and choose Create account. 2. Enter name, email and password. 3. Submit. 4. Repeat with the same email; then with a weak password and a bad email.
Expected result	A unique account is created and the user is logged in. A duplicate email gets 409 and invalid input gets 400 with a message naming the problem.

Automated tests	`test_auth::test_TC_Auth_01_register_creates_unique_account`
	`test_auth::test_TC_Auth_01b_register_validates_input`
	`test_part3_additions::test_TC_Auth_01c_name_longer_than_100_characters_refused`

System validation	SV-01, SV-02
Result	**PASS**

Test Case ID	TC-Auth-02: Log in with valid credentials
Requirement	TMA-F-002
Module / Priority	Authentication / High
Preconditions	A registered, active account exists.
Test data	Registered email (any letter case) and password.
Steps	1. Open the app. 2. Enter email and password. 3. Submit.
Expected result	User is authenticated and reaches the task dashboard. The token works on protected routes.

Automated tests	`test_auth::test_TC_Auth_02_valid_login_returns_session_token`
	`test_auth::test_login_email_is_case_insensitive`
	`test_sessions::test_TC_Auth_02b_token_gives_access_to_protected_routes`
	`test_sessions::test_update_profile`

System validation	SV-05
Result	**PASS**

Test Case ID	TC-Auth-03: Log in with invalid credentials
Requirement	TMA-F-003
Module / Priority	Authentication / High
Preconditions	A registered account exists.
Test data	Wrong password; unknown email; six failures in a row.
Steps	1. Log in with a wrong password. 2. Log in with an unknown email. 3. Fail five times, then try a sixth time and then the correct password.
Expected result	Rejected with 'Invalid email or password' and no session, the same message in both cases. After 5 failures in 15 minutes login returns 429, even with the correct password.

Automated tests	`test_auth::test_login_requires_both_fields`
	`test_invalid_login::test_TC_Auth_03_invalid_credentials_rejected`
	`test_invalid_login::test_unknown_email_still_checks_a_password_hash`
	`test_security_logging::test_TC_Auth_03b_repeated_failures_are_rate_limited`
	`test_security_logging::test_lockout_ends_when_the_window_passes`

System validation	SV-04, SV-32
Result	**PASS**

Test Case ID	TC-Auth-04: Log out
Requirement	TMA-F-004
Module / Priority	Authentication / High
Preconditions	User is logged in.
Test data	The session token in use.
Steps	1. Choose Log out. 2. Call a protected page or API with the old token.
Expected result	Session ends, the login form is shown, and the old token gets 401.
Automated tests	`test_sessions::test_TC_Auth_04_logout_terminates_session`
System validation	SV-03
Result	**PASS**

Test Case ID	TC-Task-01: Create a task with a title and optional description
Requirement	TMA-F-010
Module / Priority	Task / High
Preconditions	User is logged in.
Test data	Title up to 200 characters; description up to 2000.
Steps	1. Choose New task. 2. Save with an empty title. 3. Enter a title and description. 4. Save.
Expected result	The empty title is refused with a clear message. The valid task is saved and appears in the list.

Automated tests	`test_part3_additions::test_TC_Task_01c_description_must_be_text_of_limited_length`
	`test_part3_additions::test_TC_Task_01d_plain_task_row_falls_back_to_the_stored_status`
	`test_tasks::test_TC_Task_01_create_task_appears_in_list`
	`test_tasks::test_TC_Task_01b_create_requires_title`

System validation	SV-06
Result	**PASS**

Test Case ID	TC-Task-02: Edit an existing task
Requirement	TMA-F-011
Module / Priority	Task / High
Preconditions	User owns a task.
Test data	New title and description; one invalid field.
Steps	1. Choose Edit on a task. 2. Change the fields. 3. Save. 4. Send an update with one invalid field.
Expected result	Updated information is saved and displayed. An invalid or unknown field rejects the whole update. A save with no real change writes no history.

Automated tests	`test_part3_additions::test_TC_Task_02c_patch_without_real_change_writes_nothing`
	`test_tasks::test_TC_Task_02_edit_task`
	`test_tasks::test_TC_Task_02b_invalid_field_rejects_whole_update`
	`test_tasks::test_missing_task_returns_404`
	`test_tasks::test_unknown_fields_rejected`

System validation	SV-10
Result	**PASS**

Test Case ID	TC-Task-03: Delete a task after confirmation
Requirement	TMA-F-012
Module / Priority	Task / High
Preconditions	User owns a task.
Test data	An existing task.
Steps	1. Choose Delete. 2. Choose Keep task. 3. Choose Delete again, then Delete task.
Expected result	Keep task leaves it in place. Confirming removes it from active views and the database, with its reminders and history.
Automated tests	`test_task_lifecycle::test_TC_Task_03_delete_task` `test_task_lifecycle::test_delete_missing_task_returns_404`
System validation	SV-18
Result	**PASS**

Test Case ID	TC-Task-04: Mark a task complete and reopen it
Requirement	TMA-F-013
Module / Priority	Task / High
Preconditions	User owns an active task.
Test data	An active task.
Steps	1. Choose Complete. 2. Choose Reopen.
Expected result	Status moves between active and completed in the list. Completing twice is harmless.
Automated tests	`test_task_lifecycle::test_TC_Task_04_complete_and_reopen` `test_task_lifecycle::test_completing_twice_is_harmless`
System validation	SV-13
Result	**PASS**

Test Case ID	TC-Sec-01: Attempt to access another user's task
Requirement	TMA-F-014, TMA-SR-003
Module / Priority	Access Control / High
Preconditions	Two user accounts; user A owns a task.
Test data	User B's token; user A's task id; HTML in a task title.
Steps	1. As B, read, edit, delete, complete and add a reminder to A's task, and read its history. 2. Call each protected API route with no token. 3. As a normal user, call an admin route.
Expected result	Access denied (403 / 401), the task is unchanged, ACCESS_DENIED is logged, and B's list stays empty.

Automated tests	`test_access_control::test_TC_Sec_01_cannot_access_another_users_task`
	`test_authorization::test_TC_Sec_01b_all_api_routes_require_login`
	`test_authorization::test_public_endpoints_stay_public`
	`test_authorization::test_admin_required_checks_role`
	`test_data_protection::test_html_in_task_text_is_stored_as_plain_text`
	`test_search::test_search_never_returns_other_users_tasks`
	`test_tasks::test_task_routes_require_login`

System validation	SV-21, SV-22
Result	**PASS**

Test Case ID	TC-Sec-02: Inspect stored password data and data protection
Requirement	TMA-SR-002, TMA-NF-003
Module / Priority	Authentication / Security / High
Preconditions	Accounts exist.
Test data	Two users with one password; SQL injection strings.
Steps	1. Inspect the users table and the raw database files. 2. Search for the password and token. 3. Check response headers. 4. Send SQL injection strings to login, search, filters and sort.
Expected result	Only salted scrypt hashes, different for each user. No plaintext password or raw token. Security headers present. Injection strings have no effect.

Automated tests	`test_data_protection::test_TC_Sec_02b_security_headers_present`
	`test_data_protection::test_api_responses_are_never_cached`
	`test_data_protection::test_password_hash_never_leaves_the_server`
	`test_data_protection::test_search_input_cannot_inject_sql`
	`test_passwords::test_TC_Sec_02_passwords_stored_only_as_hashes`
	`test_passwords::test_production_uses_scrypt`
	`test_passwords::test_same_password_gets_different_hashes`

System validation	SV-29, SV-33, SV-34
Result	**PASS**

Test Case ID	TC-Sec-03: HTTPS / TLS for network communication
Requirement	TMA-SR-001
Module / Priority	Network Security / High
Preconditions	Server started with FORCE_HTTPS on.
Test data	A plain-HTTP request; an HTTPS request.
Steps	1. Send a plain-HTTP request. 2. Start with run.py --https and connect over TLS. 3. Read the protocol version and the HSTS header.
Expected result	HTTP gets 301 to https://. TLS 1.2 or later is negotiated. HSTS is sent.

Automated tests	`test_https::test_TC_Sec_03_http_redirected_to_https_with_hsts`
	`test_https::test_login_over_plain_http_never_reaches_the_app`
	`test_https::test_proxy_header_trusted_only_when_configured`
	`test_https::test_forwarded_https_accepted`
	`test_https::test_production_forces_https_development_does_not`

System validation	SV-41, SV-42
Result	**PARTIAL**

Test Case ID	TC-Sec-04: Session tokens are protected and invalidated
Requirement	TMA-SR-004
Module / Priority	Session Management / High
Preconditions	User is logged in.
Test data	Forged, missing, logged-out and expired tokens.
Steps	1. Call a protected route with each kind of token. 2. Check the sessions table.
Expected result	All are refused with 401. The database holds only a SHA-256 hash of each token.

Automated tests	`test_part3_additions::test_TC_Sec_04b_revoke_all_sessions_without_a_caller_transaction`
	`test_sessions::test_TC_Sec_04_expired_and_forged_tokens_rejected`
	`test_sessions::test_deactivated_account_token_rejected`
	`test_sessions::test_logout_only_ends_that_session`
	`test_sessions::test_missing_or_wrong_scheme_rejected`

System validation	SV-30
Result	**PASS**

Test Case ID	TC-Sec-05: Security events are logged without secrets
Requirement	TMA-SR-005
Module / Priority	Security / Audit / High
Preconditions	Administrator account exists.
Test data	Failed logins with a recognisable password.
Steps	1. Fail a login, log in, log out, trigger an access denial and an admin action. 2. Read security_events and the application log.
Expected result	The events are recorded with type, email and IP. No password or token appears in either place.

Automated tests	`test_admin::test_TC_Sec_05b_admin_actions_logged`
	`test_part3_additions::test_TC_Sec_05c_admin_reads_all_events_and_bad_limit_is_refused`
	`test_security_logging::test_TC_Sec_05_event_logged_without_secrets`
	`test_security_logging::test_access_denied_is_logged`
	`test_security_logging::test_account_lifecycle_is_logged`
	`test_security_logging::test_failed_logins_logged_without_passwords`
	`test_security_logging::test_scrubber_removes_sensitive_keys`

System validation	SV-31, SV-45
Result	**PASS**

Test Case ID	TC-Val-01: Basic input validation and fuzz style checks
Requirement	Test Plan 5.1, SAD 4.4
Module / Priority	Input validation / Medium
Preconditions	User is logged in.
Test data	Broken JSON, wrong types, over-long text, impossible dates, boundary values.
Steps	1. Send each malformed request. 2. Check the status and the error format. 3. Check the application still works.
Expected result	Every malformed request gets a 4xx in the standard error format, never a 500. The application stays healthy.

Automated tests	`test_auth::test_register_rejects_non_json_body`
	`test_input_robustness::test_TC_Val_01b_huge_id_in_any_url_is_not_found`
	`test_input_robustness::test_TC_Val_01c_largest_valid_id_is_an_ordinary_not_found`
	`test_input_robustness::test_TC_Val_01d_surrogate_in_any_json_body_is_refused`
	`test_input_robustness::test_TC_Val_01e_deeply_nested_json_is_refused`
	`test_input_robustness::test_TC_Val_01f_huge_paging_values_are_refused_or_clamped`
	`test_input_robustness::test_TC_Val_01g_dates_outside_the_supported_range_are_refused`
	`test_input_robustness::test_TC_Val_01h_dates_at_the_edges_of_the_range_are_accepted`
	`test_input_robustness::test_TC_Val_01i_reminder_before_the_earliest_date_is_refused_cleanly`
	`test_input_robustness::test_TC_Val_01j_infinite_or_huge_minutes_before_is_refused`
	`test_input_robustness::test_TC_Val_01k_surrogate_in_login_and_registration_is_refused`
	`test_input_robustness::test_TC_Val_01l_valid_unicode_is_still_accepted`
	`test_input_robustness::test_TC_Val_01m_seeded_fuzz_of_every_route_gives_no_server_error`
	`test_part3_additions::test_DEF_01_huge_integer_in_url_or_query_is_not_a_server_error`
	`test_part3_additions::test_DEF_02_date_outside_the_supported_range_is_not_a_server_error`
	`test_part3_additions::test_DEF_03_infinite_minutes_before_is_not_a_server_error`
	`test_part3_additions::test_DEF_04_lone_surrogate_in_text_is_not_a_server_error`

System validation	SV-35, SV-36a, SV-36b, SV-36c, SV-36d, SV-36e, SV-36f, SV-36g, SV-36h, SV-36i, SV-36j, SV-36k, SV-36l
Result	**PASS**

Test Case ID	TC-Org-01: Assign a priority to a task
Requirement	TMA-F-020
Module / Priority	Task / Scheduling / High
Preconditions	User is logged in.
Test data	LOW, MEDIUM, HIGH; an invalid value.
Steps	1. Create a task with High priority. 2. Change the priority. 3. Send an invalid value.
Expected result	Selected priority is saved and displayed. An invalid value gets 400. The change is in the history.

Automated tests	`test_priority::test_TC_Org_01_priority_saved_and_displayed`
	`test_priority::test_invalid_priority_rejected`
	`test_priority::test_priority_change_is_recorded_in_history`

System validation	SV-07
Result	**PASS**

Test Case ID	TC-Org-02: Assign and update task status
Requirement	TMA-F-021
Module / Priority	Task / Scheduling / High
Preconditions	User owns a task.
Test data	ACTIVE, COMPLETED; a past due date.
Steps	1. Set the status to Completed and back. 2. Give an active task a past due date. 3. Try to set OVERDUE by hand.
Expected result	Status is saved and shown. A past-due active task reads Overdue automatically. Setting OVERDUE by hand gets 400.

Automated tests	`test_status::test_TC_Org_02_status_assigned_and_updated`
	`test_status::test_overdue_cannot_be_set_by_hand`
	`test_status::test_overdue_is_derived_from_the_due_date`
	`test_status::test_status_change_is_logged_in_history`

System validation	SV-12
Result	**PASS**

Test Case ID	TC-Org-03: Create or select categories / tags
Requirement	TMA-F-022
Module / Priority	Task / Scheduling / Medium
Preconditions	User is logged in.
Test data	New names; an existing name in another letter case.
Steps	1. Save a task with new category names. 2. Save another using an existing name in lower case. 3. Delete a category.
Expected result	Task is linked to the categories. Names are reused regardless of case. Deleting a category keeps its tasks.

Automated tests	`test_categories::test_TC_Org_03_create_or_select_categories`
	`test_categories::test_categories_are_private`
	`test_categories::test_deleting_a_category_keeps_its_tasks`
	`test_categories::test_edit_replaces_categories`
	`test_categories::test_invalid_categories_rejected`
	`test_part3_additions::test_TC_Org_03b_blank_category_names_are_dropped`

System validation	SV-09
Result	**PASS**

Test Case ID	TC-Org-04: Set or change a task due date
Requirement	TMA-F-023
Module / Priority	Scheduling / High
Preconditions	User owns a task.
Test data	Local date and time; invalid dates; empty value.
Steps	1. Enter a due date in local time. 2. Change it. 3. Clear it. 4. Send an invalid date.
Expected result	The date is stored as UTC and displayed correctly in local time. It can be cleared. Invalid dates get 400.

Automated tests	`test_due_dates::test_TC_Org_04_due_date_stored_and_displayed`
	`test_due_dates::test_due_date_can_be_changed_and_cleared`
	`test_due_dates::test_due_date_change_is_recorded_in_history`
	`test_due_dates::test_invalid_due_date_rejected`
	`test_part3_additions::test_TC_Org_04b_time_without_offset_is_taken_as_utc`

System validation	SV-08
Result	**PASS**

Test Case ID	TC-Org-05: Configure a reminder
Requirement	TMA-F-024
Module / Priority	Notification / Medium
Preconditions	User owns a task with a due date.
Test data	At the due time; 15, 60 or 1440 minutes before; an exact time.
Steps	1. Choose a reminder in the task form. 2. Check it is shown on the task. 3. Wait until it is due.
Expected result	Reminder is stored and fires once at the selected time. A time in the past or a relative reminder without a due date gets 400.

Automated tests	`test_notifications::test_TC_Org_05b_reminder_triggered_once_at_its_time`
	`test_reminders::test_TC_Org_05_reminder_configured_and_stored`
	`test_reminders::test_delete_reminder`
	`test_reminders::test_invalid_reminders_rejected`
	`test_reminders::test_minutes_before_needs_a_due_date`
	`test_reminders::test_minutes_before_uses_the_due_date`
	`test_reminders::test_reminders_deleted_with_their_task`

System validation	SV-11
Result	**PASS**

Test Case ID	TC-Search-01: Search tasks by title or description keyword
Requirement	TMA-F-030
Module / Priority	Search / Medium
Preconditions	User has several tasks.
Test data	Matching and non-matching keywords.
Steps	1. Type a keyword from a description. 2. Type a word that matches nothing.
Expected result	Matching tasks are returned and non-matching tasks are excluded. Other users' tasks are never returned.
Automated tests	`test_search::test_TC_Search_01_keyword_matches_title_or_description`
System validation	SV-15
Result	**PASS**

Test Case ID	TC-Search-02: Filter by status, priority and category
Requirement	TMA-F-031
Module / Priority	Search / Medium
Preconditions	User has varied tasks.
Test data	Each filter value; an invalid value.
Steps	1. Apply each filter in turn. 2. Clear the filters. 3. Send an invalid filter value.
Expected result	Only matching tasks are displayed. An invalid value gets 400.
Automated tests	`test_search::test_TC_Search_02_filters` `test_search::test_invalid_filters_rejected`
System validation	SV-16
Result	**PASS**

Test Case ID	TC-Search-03: Sort by due date, priority or status
Requirement	TMA-F-032
Module / Priority	Search / Medium
Preconditions	User has varied tasks.
Test data	Each sort field; both orders.
Steps	1. Choose each sort option. 2. Reverse the order.
Expected result	The selected order is applied. Tasks without a due date come last.
Automated tests	`test_search::test_TC_Search_03_sorting` `test_search::test_pagination`
System validation	SV-17
Result	**PASS**

Test Case ID	TC-Progress-01: Progress summary
Requirement	TMA-F-033
Module / Priority	Dashboard / Medium
Preconditions	User has active, completed and overdue tasks.
Test data	A mix of task states.
Steps	1. Open the dashboard. 2. Compare its counts with the task data. 3. Complete a task and check again.
Expected result	Counts match the underlying task data. Other users' tasks are not counted.

Automated tests	`test_dashboard::test_TC_Progress_01_counts_match_task_data`
	`test_dashboard::test_due_soon_lists_open_tasks_due_this_week`
	`test_dashboard::test_empty_dashboard`
	`test_dashboard::test_other_users_tasks_not_counted`

System validation	SV-14
Result	**PASS**

Test Case ID	TC-Notify-01: Notification generated for a due reminder
Requirement	TMA-F-040
Module / Priority	Notification / Medium
Preconditions	A reminder is about to fall due.
Test data	A reminder a few seconds away.
Steps	1. Wait for the reminder without using the app. 2. Open the app and read the badge and list. 3. Mark all as read.
Expected result	Exactly one in-app notification is created by the background thread. The badge shows it. A completed task gives no notification.

Automated tests	`test_notifications::test_TC_Notify_01_notification_generated_for_due_reminder`
	`test_notifications::test_completed_tasks_do_not_notify`
	`test_notifications::test_lazy_trigger_when_listing_notifications`
	`test_notifications::test_notifications_are_private`
	`test_notifications::test_send_reminders_cli`
	`test_part3_additions::test_TC_Notify_01b_background_thread_delivers_due_reminders`
	`test_part3_additions::test_TC_Notify_01c_background_thread_survives_a_failed_pass`
	`test_part3_additions::test_TC_Notify_01d_background_thread_pass_with_nothing_due`
	`test_part3_additions::test_TC_Notify_01e_email_not_sent_when_the_service_is_disabled`
	`test_part3_additions::test_TC_Notify_01f_email_handed_to_the_service_when_enabled`
	`test_part3_additions::test_TC_Notify_01g_email_interface_returns_whether_it_was_sent`
	`test_part3_additions::test_TC_Notify_01i_mark_all_as_read`

System validation	SV-19
Result	**PASS**

Test Case ID	TC-Notify-02: Manage notification and task preferences
Requirement	TMA-F-041
Module / Priority	Notification / Medium
Preconditions	User is logged in.
Test data	In-app off; default priority and sort.
Steps	1. Open Preferences and change the values. 2. Save. 3. Create a task. 4. Let a reminder fire.
Expected result	Changes are saved and applied: the new defaults are used and no in-app notification is created while it is off.

Automated tests	`test_notifications::test_TC_Notify_02_preferences_apply_to_later_notifications`
	`test_notifications::test_defaults_before_anything_is_saved`
	`test_notifications::test_invalid_preferences_rejected`
	`test_notifications::test_task_preferences_apply_to_new_tasks_and_the_list`

System validation	SV-20
Result	**PASS**

Test Case ID	TC-Admin-01: Administrator user management
Requirement	TMA-F-042
Module / Priority	Administration / Medium
Preconditions	An administrator made with flask create-admin.
Test data	Admin, normal and anonymous users.
Steps	1. As a normal user and anonymously, call an admin route. 2. As admin, deactivate a user. 3. Try to use that user's session and to log in.
Expected result	Only administrators reach admin functions. A deactivated user is logged out at once and cannot log in. Admins cannot lock themselves out.

Automated tests	`test_admin::test_TC_Admin_01_only_admins_can_manage_users`
	`test_admin::test_admin_cannot_lock_themselves_out`
	`test_admin::test_create_admin_cli`
	`test_admin::test_deactivation_logs_the_user_out_and_blocks_login`
	`test_admin::test_invalid_admin_updates_rejected`
	`test_admin::test_promote_and_delete_user`
	`test_admin::test_security_events_review`
	`test_invalid_login::test_deactivated_account_cannot_log_in`
	`test_part3_additions::test_TC_Admin_01b_create_admin_command_reports_invalid_input`

System validation	SV-23
Result	**PASS**

Test Case ID	TC-Perf-01: Load the dashboard under normal load
Requirement	TMA-NF-001
Module / Priority	Application / UI / High
Preconditions	A running server with seeded task data.
Test data	10 concurrent clients, 500 tasks per user (assumed, as the SRS does not define normal load).
Steps	1. Send the requests. 2. Compute the 95th percentile. 3. Repeat for search and filter.
Expected result	95th percentile response time is 3 seconds or less, with no failed requests.

Automated tests	`test_performance::test_TC_Perf_01_dashboard_p95_under_3_seconds`
	`test_performance::test_task_list_with_300_tasks_is_fast`
	`test_performance::test_task_queries_use_indexes`

System validation	SV-38, SV-39
Result	**PASS**

Test Case ID	TC-Reliability-01: System availability
Requirement	TMA-NF-002
Module / Priority	Infrastructure / Ops / High
Preconditions	A running server.
Test data	Health endpoint; abrupt kill and restart.
Steps	1. Call /api/health. 2. Kill the server. 3. Restart it and check again.
Expected result	Health reports status (503 when the database fails) and the service recovers. Monthly uptime of at least 99.5%.

Automated tests	`test_health::test_TC_Reliability_01_health_endpoint`
	`test_health::test_app_recovers_after_restart`
	`test_health::test_health_reports_database_failure`
	`test_health::test_health_reveals_nothing_private`

System validation	SV-44
Result	**PARTIAL**

Test Case ID	TC-Integrity-01: Data consistency across operations
Requirement	TMA-NF-004
Module / Priority	Database / Task / High
Preconditions	A running server.
Test data	Repeated create, update, complete and delete; 200 concurrent creates; a kill.
Steps	1. Repeat the operations and compare with the expected state. 2. Force a failure inside a save. 3. Create tasks concurrently. 4. Kill the server and restart it.
Expected result	Database state is as expected. A failed save leaves nothing behind. Confirmed changes survive a crash.

Automated tests	`test_integrity::test_TC_Integrity_01_repeated_crud_preserves_state`
	`test_integrity::test_crash_while_saving_a_task_leaves_nothing_behind`
	`test_integrity::test_failed_transaction_rolls_back`
	`test_integrity::test_invalid_edit_changes_nothing`
	`test_part3_additions::test_TC_Integrity_01b_init_db_command_is_safe_to_repeat`

System validation	SV-37, SV-40, SV-43
Result	**PASS**

Test Case ID	TC-UX-01: Responsive and accessible user interface
Requirement	TMA-NF-005
Module / Priority	UI / Medium
Preconditions	A running server; a browser.
Test data	360, 390, 768 and 1280 px widths; dark mode; keyboard only.
Steps	1. Check each width for sideways scroll. 2. Use the main flow with the keyboard. 3. Check accessible names and contrast. 4. Switch to dark mode. 5. Go offline and try to save.
Expected result	No sideways scroll or clipped controls. Keyboard flow works with a visible focus. Controls are named. Contrast meets WCAG AA. Network failures give a clear message.

Automated tests	`test_part3_additions::test_TC_UX_01b_unknown_page_keeps_the_normal_not_found_page`
	`test_ux::test_TC_UX_01_responsive_and_every_control_is_labelled`
	`test_ux::test_dialogs_and_tables_are_named`
	`test_ux::test_static_assets_served`
	`test_ux::test_status_messages_are_announced_and_skip_link_exists`
	`test_ux::test_text_colours_meet_wcag_aa_contrast_in_both_themes`

System validation	SV-24, SV-25, SV-26, SV-27, SV-28
Result	**PASS**

4. Supporting tests not tied to one case

These check the application skeleton shared by all modules.

- `test_app::test_database_schema_created`
- `test_app::test_index_page_served`
- `test_app::test_unexpected_errors_return_clean_json`
- `test_app::test_unknown_api_route_returns_json_404`